Experiment No:6
Time Series Analysis of Website Traffic

Objective:

To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Dataset: A dataset containing daily website traffic data with columns such as Date, Page_Views, Unique_Visitors, and Bounce_Rate.

Methodology:

1. Load and Clean Data:

o Import the dataset and parse the Date column as a datetime object.
o Handle missing values by interpolation.

2. Time Series Decomposition:

o Decompose the time series data of Page_Views to identify trend, seasonality, and residuals.

3. Visualization:

o Plot the time series data for Page_Views, Unique_Visitors, and Bounce_Rate.
o Use moving averages to smooth the time series and highlight trends.
o Create seasonal plots to visualize patterns over time.

4. Anomaly Detection:

o Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset
data = pd.read_csv("/content/website_wata_with_date.csv",
parse_dates=['Date'], index_col='Date')

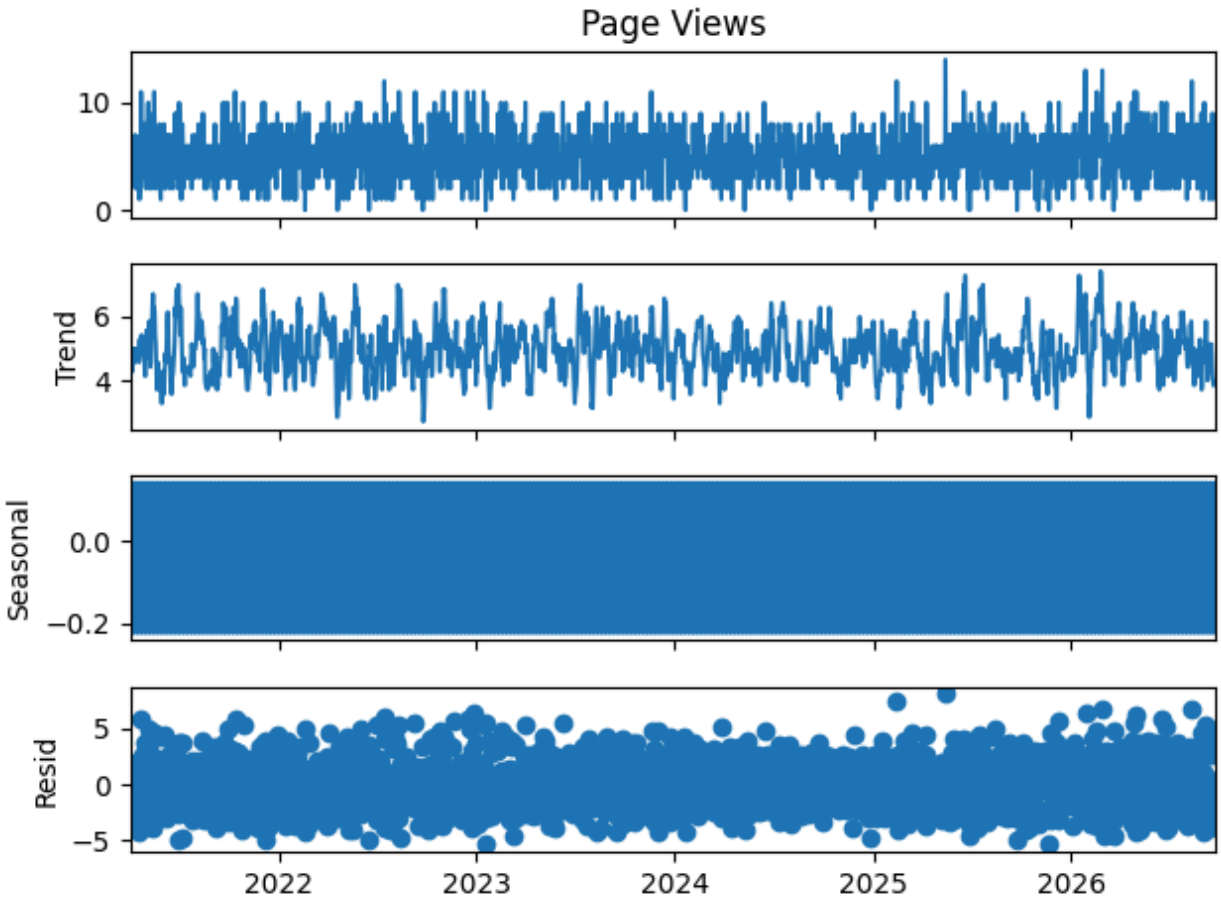
# Time Series Decomposition
decomposition = seasonal_decompose(data['Page Views'], model='additive')
decomposition.plot()
```

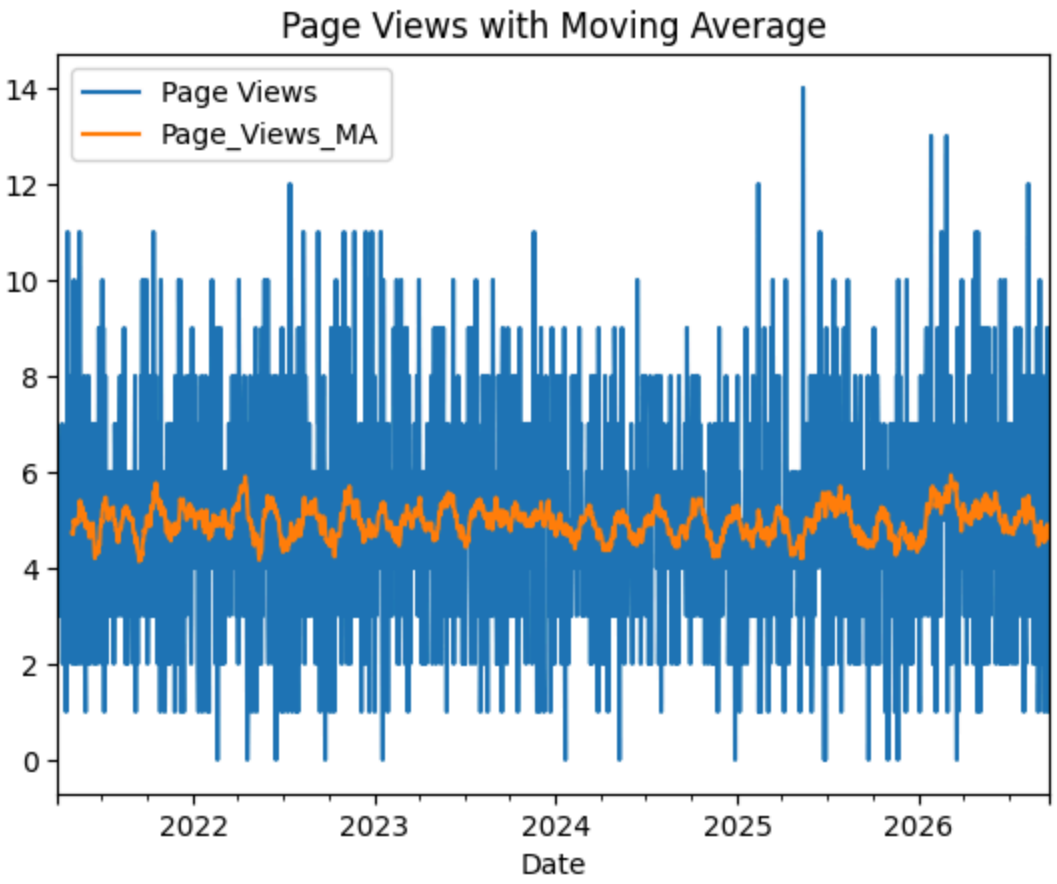
```
plt.show()

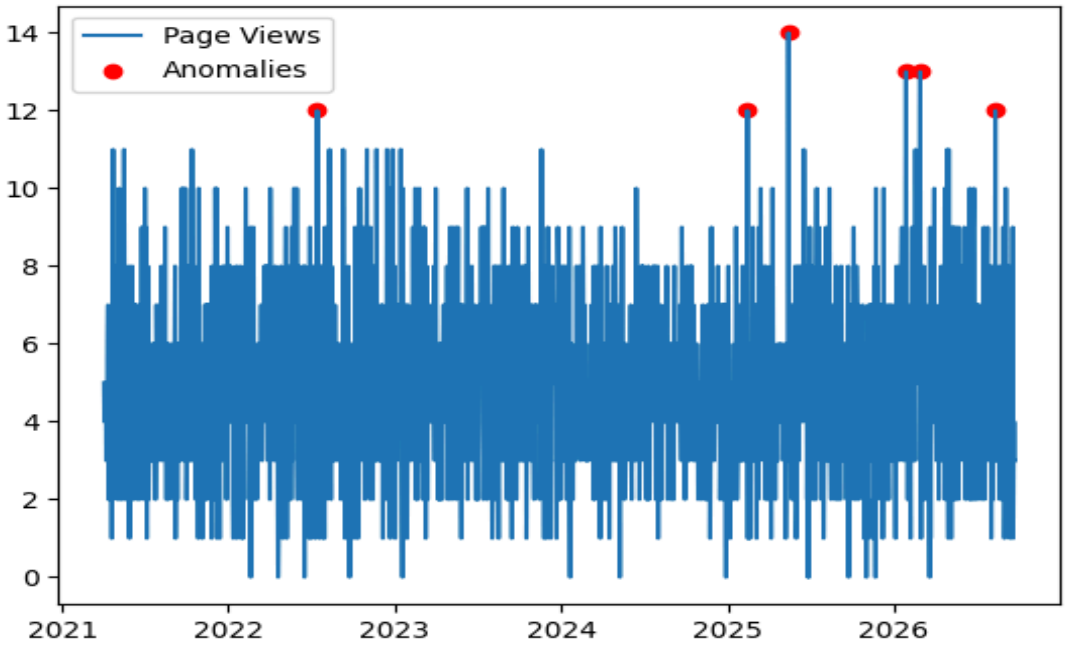
# Moving Average
data['Page_Views_MA'] = data['Page Views'].rolling(window=30).mean()
data[['Page Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()

# Anomaly Detection
data['Page_Views_Z'] = (data['Page Views'] - data['Page Views'].mean())
/data['Page Views'].std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page Views'], color='red',
label='Anomalies')
plt.legend()
plt.show()
```

OUTPUT:


Page Views
10
0
Trend
6
4
Seasonal
0.0
−0.2
Resid
5
0
−5
2022
2023
2024
2025
2026


Page Views with Moving Average
Page Views
Page_Views_MA
14
12
10
8
6
4
2
0
2022
2023
2024
2025
2026
Date


Page Views
Anomalies
14
12
10
8
6
4
2
0
2021
2022
2023
2024
2025
2026

RESULT:

The website is visualized using a line plot,showing the variations and sudden increases in page views